

Project Initialization and Planning Phase

Date	06 July 2026
Team ID	XXXXXXXXXX
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to enhance the online learning experience by integrating artificial intelligence and emotion detection. The system analyzes a student's facial expressions in real time to identify their emotional state and recommends personalized learning resources based on the detected emotion. This approach improves student engagement, concentration, and academic performance while providing a more adaptive learning environment.

Project Overview	
Objective	The primary objective is to develop an AI-powered learning support system that detects students' emotions using facial expression recognition and delivers personalized learning recommendations to improve understanding, engagement, and academic performance.
Scope	The project focuses on real-time emotion detection through a webcam, AI-based emotion classification, personalized study material recommendations, and student progress tracking using an interactive web application.
Problem Statement	
Description	Traditional online learning platforms provide the same learning content to every student without considering their emotional state. Stress, frustration, and lack of motivation often reduce concentration and learning efficiency.
Impact	Addressing these challenges will improve student engagement, increase learning effectiveness, enhance academic performance, and create a more personalized and supportive learning experience.
Proposed Solution	
Approach	The system captures facial expressions through a webcam, detects the student's emotion using a deep learning model, and recommends appropriate learning resources, study strategies, and practice activities according to the detected emotional state.

Key Features	☐ Real-time facial emotion detection using AI. ☐ Personalized learning resource recommendations. ☐ Subject-wise study assistance. ☐ Interactive Streamlit web application. ☐ Student progress and emotion tracking. ☐ AI-powered learning support for improved engagement.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU for model inference	Intel i5/Ryzen 5 or higher (GPU optional)
Memory	RAM specifications	Min 8 GB
Storage	Application, datasets and models	10–20 GB SSD
Software		
Frameworks	Python frameworks	Streamlit
Libraries	AI & Data Processing	TensorFlow/Keras, OpenCV, NumPy, Pandas, Matplotlib
Development Environment	IDE	Visual Studio Code / PyCharm
Data		
Data	Facial Emotion Recognition Dataset	FER-2013 (CSV) or equivalent facial emotion dataset